

Hypo- and hypermobility

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Introduction

Joint mobility is a fundamental prerequisite to human movement and motor development. Commonly measured as range of motion, joint mobility is the controlled motion of a bony lever on articular surfaces (Berryman Reese and Bandy, 2017). Cardinal planes of motion and axes of rotation (frontal, sagittal, transverse) further standardize these motions. While range of motion is variable among individuals, there are suggested norms required for movement efficiency (Soucie et al., 2011). Deviation from these normative values is characterized as either hypomobility or hypermobility. Identifying and addressing impairments in joint mobility are essential components of managing aberrant movement patterns (Keer and Simmonds, 2011).

Hypomobile individuals frequently present to yoga and Pilates classes to improve their flexibility and overall physical condition but, dependent upon the underlying cause of their hypomobility, they may pose specific challenges not only for yoga and Pilates instructors, but also for other bodyworkers, physiotherapists, Feldenkrais and Alexander teachers, and neuromuscular therapists. In this chapter, we use a broad definition of “bodywork” to encompass all these and other disciplines. This chapter provides an overview of specific hypomobile syndromes and associated issues that bodyworkers should at least be familiar with. On the other hand, hypermobile persons are commonly thought of as excellent

candidates for Pilates and yoga, as often they have no difficulty assuming difficult poses and stretching much further than the average yoga student. Yet hypermobility features its own specific challenges. Persons with Ehlers-Danlos Syndrome, for example, may have such loose connective tissue and lack fascial stability that maintaining certain poses or performing core training may be difficult.

Hypomobility

Hypomobility can be defined as decreased joint mobility due to inherited or acquired structural changes in the joint, joint capsule or other surrounding tissues, or because of functional alterations of the control system. It is characterized by decreased range of motion, altered movement activation patterns, shortened muscles and fascia, and possible swelling. There continues to be a paucity of studies on hypomobility and of papers describing the prevalence of hypomobility occurring at different joints. Persons who are just a bit out of shape and who have developed tight muscles and restricted fascia can do very well with bodywork interventions. Others, with more complex inherited joint hypomobility, may require significant modifications and individualized sessions addressing their unique problems.

Inherited joint hypomobility

Inherited joint hypomobility is a very complex issue. The prevalence of inherited conditions

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featuring joint hypomobility is probably very low, but it cannot be determined accurately without objective data. This section provides much detailed information about several inherited disorders in the hope that familiarity with these disorders will improve the quality of life for participants, and perhaps provide new business opportunities for “disease-specific” yoga classes, or Pilates mat work. Few yoga and Pilates instructors may be familiar with these disorders, and the possible benefits of bodywork have hardly been explored. Recognizing that persons with these disorders may not often present to Pilates, yoga or bodywork, we recommend consulting this section as necessary.

Inherited joint hypomobility can be related to a variety of genetic diagnoses, such as arthrogryposis multiplex congenita, which is defined by the presence of congenital non-progressive contractures across two or more major joints (Valdes-Flores et al., 2016). The prevalence of arthrogryposis multiplex congenita is 1:3000–5000 newborns (Valdes-Flores et al., 2016). The disorder is not only linked to nervous system disorders, such as focal or generalized anterior horn cell deficiency, and structural brain disorder or damage, but can also be associated with a wide range of other diagnoses, such as dysplasias (diastrophic, arastremmatic, Kniest, metatropic, and campomelic dysplasia), Schwartz syndrome, fetal alcohol syndrome with synostoses, osteogenesis imperfecta, Emery–Dreifuss muscular dystrophy, antecubital webbing syndrome, various pterygium syndromes (i.e. popliteal and multiple pterygium syndrome), and Freeman Sheldon syndrome, among others (Mennen et al., 2005). Because of the wide range of conditions linked to arthrogryposis, there are multiple possible genetic mutations and signaling pathways. Several researchers implicated mutations in X-linked zinc-finger gene ZC4H2 in

the etiology of arthrogryposis multiplex congenita (Okubo et al., 2018; Kondo et al., 2018), while others identified RYR1 mutations (Brackmann et al., 2018), TTN (Fernandez-Marmiesse et al., 2017), LG14 (Xue et al., 2017) and BICD2 mutations (Ravenscroft et al., 2016), among others. Three consanguineous families presented with different homozygous mutations in GPR126, which is required to encode G-protein-coupled receptor 126, a critical step in peripheral nervous system axonal myelination (Ravenscroft et al., 2015). In other words, the diagnosis of arthrogryposis can involve a wide range of genetic variations and, therefore, rather different presentations.

It is not necessary for bodyworkers, Pilates and yoga instructors to fully comprehend the many complexities of arthrogryposis, but they may be able to contribute to a better quality of life and at least temporary improvements in range of motion through active participation in regular individualized exercise programs. Bodywork is often offered in a non-medical environment, which may be an appealing alternative for this population. Little is known about the specific fascial restrictions in persons with arthrogryposis. It is, however, possible that the contractures feature extreme fascial webbing, whereby the main nerves and blood vessels are also shortened. In extreme cases, webbing of the knee joint may require extensive surgical release of the webbing and even shortening of the femur (Saleh et al., 1989; Ponten, 2015). In such extreme cases, bodywork approaches have limited value.

Other inherited diseases featuring joint hypomobility include congenital talipes equinovarus (clubfoot), idiopathic toe walking and symptomatic generalized hypomobility. The incidence of congenital talipes equinovarus ranges from 1–3 cases per 1000 pregnancies, with males twice as frequently affected as females.

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It is less common in Asian populations, but more common among Polynesians and the Maoris in New Zealand, with an incidence of 6.5–7 per 1000 births (Drvaric et al., 1989; Chapman et al., 2000). The incidence for Indians and Pakistanis living in the UK was 7.8 per 1000 (Leck and Lancashire, 1995). One of the most used treatments is the Ponseti method, which involves serial casting followed by surgical correction for resistant deformity (Khanna and Vaishya, 2017). The method is widely used throughout the world and it is still considered the gold standard, although recently Andreoli et al. (2014) described a case of a newborn treated with serial casting combined with osteopathic manipulative treatment without having to resort to surgery. Considering that osteopathic manipulations were beneficial, perhaps there are also opportunities for modified yoga interventions, even although that would require some unorthodox, out-of-the-box thinking.

While the management of congenital talipes equinovarus is well established, not much is definitively known about the genetic predisposition and the signaling pathways. The syndrome is likely to be the result of a variety of environmental and genetic factors. Some older papers suspected an autosomal recessive inheritance, but, more likely, the development of clubfoot may be a result of one major gene effect involving interactions with additional polygenes and environmental factors (Dietz, 2002; Rebbeck et al., 1993). Basit and Khoshhal (2018) summarized the current state of affairs in a recent review. They described that genes involved in coding for contractile proteins of skeletal myofibers may be involved, but while there is no clearly identified genetic deficit, variants in *TBX4*, *PITX1*, *HOXA*, *HOXC* and *HOXD* clusters genes, *NAT2* and others seem to play a significant role. The importance of the developmental transcriptional

pathway, known as the *PITX1-TBX4* pathway, was recently confirmed (Dobbs and Gurnett, 2012). *TBX4* and *PITX1* are expressed in the hindlimb, which helps explain the foot phenotype observed with transcription mutations of these factors. To break this down a bit more, problems in the feet are more common than in other areas, which is explained by a genetic predisposition. Several clubfoot-like deformities are associated with craniocarpotarsal and diastrophic dysplasia, Smith-Lemli-Opitz syndrome, Pierre Robin syndrome, amyoplasia, myelomeningocele, congenital constriction bands, intrauterine poisoning by aminopterin or d-tubocurarine, some Larsen syndromes, distal arthrogryposis, and various chromosomal abnormalities (Dietz, 2002).

Idiopathic toe walking is a diagnosis by exclusion, presenting in children aged older than 3 years who exhibit an abnormal gait pattern with excessive forefoot weight bearing. It may be associated with cerebral palsy (Barber et al., 2012), Duchenne muscular dystrophy (Hyde et al., 2000), McArdle's disease (Pomarino et al., 2018), and even attention deficit and hyperactivity disorder (Soto Insuga et al., 2018). In a minority of cases a familial predisposition is suspected (Pomarino et al., 2016). It could be related to a congenital short Achilles tendon (Shulman et al., 1997), but it is also conceivable that Achilles tendon shortening develops as a result of persistent toe walking, especially since at birth these children typically do not present with contractures (Brunt et al., 2004). Others have described a link with sensory dysfunction (Montgomery and Gauger, 1978).

Symptomatic generalized hypomobility has been described as a dysfunction with a systemic decrease in range of motion, an increased stiffness of joint ligaments and musculo-tendinous structures, lower bone density, excessive

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pyridinoline cross-links, and keloid scarring (Engelbert et al., 2004). There are no additional data available about symptomatic generalized hypomobility, but yoga and Pilates are likely beneficial for this population.

Acquired joint hypomobility

Acquired hypomobility can be due to a wide range of causes, such as scar tissue, kinesiophobia (fear of movement), increased age, psychological factors, inactivity and a sedentary lifestyle, and certain medical conditions, such as scleroderma, osteoarthritis, fractures, stiff person syndrome, Parkinson syndrome, cerebral palsy and spasticity, and complex regional pain syndrome, among others. Unlike for most inherited joint mobility syndromes, the role yoga and Pilates can play in the overall well-being of persons with acquired joint hypomobility is easier to grasp. Fascial restrictions associated with these disorders can be addressed by specific stretching routines and improved whole-body stability and flexibility exercises.

Within physiotherapy, chiropractic and bodywork disciplines, there seems to be a tendency to approach hypomobility from a primary biomechanical perspective, which, in many cases, may be counterproductive. For example, individuals suffering from depression may present with functional hypomobility of the shoulder joints (Okely et al., 2019). Clinicians using a strict biomechanical approach would likely miss the subtleties of the more complex presentation of depressed patients. Silva et al. (2016) described bilateral joint hypomobility of the subtalar joints in women diagnosed with fibromyalgia, whereas Brown et al. (2016) found that patients with higher levels of kinesiophobia had a greater range of motion deficits following a total knee arthroplasty. Bodyworkers should implement

a comprehensive approach in dealing with persons with hypomobility issues.

With regards to altered control systems, hypomobility and hypokinetic symptoms are common in patients suffering from Parkinson syndrome. This may be linked to dysfunction of the basal ganglia, especially the subthalamic nucleus (STN) (Di Matteo et al., 2008). A decreased STN function results in hypokinetic symptoms and joint stiffness or hypomobility, commonly observed in patients with Parkinson. The innervation of the STN is strongly influenced by serotonin, which in turn is mediated by intra-STN serotonin receptors, such as the 5-HT_{1A}, 5-HT_{2A} and 5-HT_{2C} receptors (Xiang et al., 2005). The peripheral administration of the 5-HT_{2A/C} agonist, 2,5-dimethoxy-4-iodoamphetamine (DOI), leads to a significant reduction of mobility (Hameleers et al., 2007). Increasing STN neuronal activity results in a decrease in mobility, which can be reversed by high frequency brain stimulation (HFS). Even although there are complex underlying reasons for the noted hypomobility in Parkinson syndrome, regular participation in Pilates, yoga and other forms of bodywork may possibly slow down the degree of dysfunction and provide at least temporary improved mobility.

Another diagnosis whereby a strict biomechanical orientation is not useful is a central nervous system autoimmune disease known as the stiff-person syndrome (Ciccotto et al., 2013). The disorder has also been observed in infants, where it often is described as infantile hyperexplexia, as well as in children and adolescents (Tracy and McKeon, 2015). The syndrome features muscular rigidity and painful muscle spasms primarily in the proximal limb and axial muscles, including the paraspinal muscles. The disease is associated with auto-antibodies to glutamic acid

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decarboxylase, which can impair the synthesis of one of the main inhibitory neurotransmitters of the brain known as gamma-aminobutyric acid (GABA). Stiff-person syndrome must be differentiated from other rare disorders such as subacute progressive encephalomyelitis (Hutchinson et al., 2008), paraneoplastic syndrome (Sweeney and Howe, 2016), and acquired neuromyotonia (Isaacs Syndrome) (Rana et al., 2012) to assure proper individualized medical management. As with Parkinson syndrome, individuals with stiff-person syndrome may benefit from individualized bodywork.

Persons with stiff-person syndrome typically do not have pyramidal and extrapyramidal dysfunction. The pyramidal system controls voluntary movements and dysfunction of the system may lead to paralysis, for example, when someone suffers from a cerebrovascular accident, or to increased reflexes and spasticity (Tranchant et al., 2017). The extrapyramidal system controls automatic movement patterns and dysfunction of that system may cause involuntary movements such as tics, tremors, myoclonus, chorea and dystonia, among others (Cotelli et al., 2018). Examples of extrapyramidal syndromes are progressive supranuclear palsy, Parkinson's disease and corticobasal syndrome.

Range of motion

It is obvious that joint hypomobility can have far reaching consequences. For example, hypomobility of the temporomandibular joints may lead to airway obstruction, difficulty with mastication, eating and other oral functions, impaired speech and language development, and mandibular growth restrictions of the craniomandibular complex (Costello and Edwards, 2005; Sahdev et al., 2019). Hypomobility of the finger joints may negatively impact manual dexterity and precision grip force control (Campos Cde et al., 2013).

Knee, ankle and foot hypomobility can cause significant impairments of gait and overall mobility (Delafontaine et al., 2015), whereas spinal hypomobility may reduce overall functionality and mobility (Alqhtani et al., 2015; Mustur et al., 2009).

There is disagreement, however, whether restricted range of motion as the prime feature of hypomobility can be assessed accurately in the clinical setting, particularly when it concerns segmental spinal mobility. Performing a manual posterior-anterior cervical gliding test for the mid-cervical spine was comparable with the results of dynamic radiographic assessment for the diagnosis of intervertebral hypomobility (Rey-Eiriz et al., 2010). To determine whether a certain assessment method is valid and reliable, two or more examiners should reach acceptable levels of agreement regarding the findings, which is referred to as interrater reliability. Intrarater reliability refers to whether the same examiner consistently has the same findings or scores on a test. Two examiners reached good interrater reliability with palpation for stiffness in the cervical, thoracic and lumbar spine (Holt et al., 2018), which means that they reached the same conclusions following an examination. On the contrary, the interrater and intrarater reliability of passive physiological accessory movement assessment of the lumbar spine was poor when performed by novice physical therapists (Deore and May, 2012). Three experienced physical therapists also had poor interrater reliability for testing segmental mobility of the lumbar spine (Johansson, 2006), which was similar to the findings of Schneider et al. (2008), who observed poor interrater reliability for the manual assessment of lumbar segmental motion restrictions. They did establish that palpation to provoke pain responses was more reliable. Kopenhagen et al. (2014) concluded that the sensitivity and specificity estimates of

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judgments of hypomobility were low with no correlation between manual assessments and measures of stiffness. A systematic review found limited evidence in support of reliability and validity of clinical tests to assess cervical mobility, head posture and location of pain, and in adults with neck pain and associated disorders (Lemeunier et al., 2018). In a study of the interrater reliability of the cervicothoracic and shoulder physical examination in patients with a primary complaint of shoulder pain, most cervical and shoulder range of motion measurements had variable reliability co-efficients ranging from slight to substantial agreement; however, horizontal shoulder adduction had poor reliability (Burns et al., 2016). It is of interest that Abbott et al. (2005) established that manual clinical examination procedures do have moderate validity for detecting segmental motion abnormality, which is in line with the findings of Haneline et al. (2008), who identified several studies reporting high levels of reproducibility. Furthermore, Widerström et al. (2012) found substantial interrater reliability for pain modulation, stabilization exercise and training, but not for manipulations, which was poor.

To overcome the limitations of manual range of motion measurements, the use of three-dimensional computerized motion analysis systems may provide more accuracy and reduce inherent human errors (Karakostas et al., 2018). The most common motion analysis systems can be divided into motion sensor systems and high-speed or infrared camera systems with reflective markers. Both systems are designed to not only measure range of motion, but also to provide detailed kinetic and kinematic analyses. The utility of motion analysis system is well established for the lower extremities, but much less so for the spine and upper extremities (Piche et al., 2007; Karakostas et al., 2018). There are also other issues, such as the high costs of computerized

systems, the lack of established normative data, and the inherent variability in complexity of joint structures, making standardized assessment methods difficult.

Independent of the methods used to determine range of motion, clinicians need to consider regional interdependences (McDevitt et al., 2015), for example, the mobility of the thoracic spine influences the position of the scapula and, therefore, the mobility of the upper extremity (Theodoridis and Ruston, 2002). Patients with non-specific chronic low back pain may also present with altered positions of the scapula, which in turn may trigger movement dysfunctions of the upper extremities (Taghizadeh et al., 2017). In addition, movement dysfunction of the thoracic spine has been linked to pain in the shoulder and upper arm, which by itself will likely limit the active range of motion of the shoulder joints (Berglund et al., 2008). In other words, measuring shoulder elevation without taking the mobility of the thoracic spine into account may give inaccurate results, which inevitably would lead to less than optimal treatment strategies (Puentedura and Cleland, 2015). Similarly, cervical dysfunction may be linked to unilateral lateral epicondylalgia (tennis elbow) (Coombes et al., 2014). There are several other interdependences in the body, partially because of the interconnectedness of various fascial layers. Pilates, yoga and Feldenkrais are excellent whole-body approaches to address the interconnectedness of restrictions in range of motion. There is no question that the many fascial connections between different body regions play a key role in the regional interdependences.

Physiotherapy

Many patients with hypomobility consult with physiotherapists. As mentioned before, clinicians who adhere to a strict biomechanical approach may be too restrictive and, therefore,

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it is encouraging that within the field of physiotherapy other approaches have been developed. Sahrman et al. (2017) developed a comprehensive kinesiopathologic model of “movement system impairment syndromes”, which assumes that repeated movement patterns and sustained suboptimal joint positions may lead to the development of pathology and pathoanatomical changes in tissues and joint structures. The model incorporates “the musculoskeletal system as the effector of movement, the nervous system as the regulator of movement, and the cardiovascular, pulmonary, and endocrine systems as providing support for the other systems, but that also are affected by movement”. Rabey et al. (2017) concluded that manual skills, including range of motion measurements such as the assessment of passive accessory and physiological movements, are valid tools when used judiciously in the context of an individualized, multidimensional approach based on contemporary pain science and patient-centered values. Of course, bodyworkers should approach their hypomobile clients from this contextual point of view.

From a neurophysiological perspective, segmental joint hypomobility may occur due to dysfunction of neurophysiological processing, which often is linked to the experience of pain with movement and a concurrent increase in muscle tone. The interaction between extra-articular and intra-articular aspects of movement patterns plays a significant role in joint mobility, which involves both peripheral and central neural mechanisms, passive structures, including ligaments, joint capsules and intra-articular discs, and active controls from muscles and to a lesser degree from fascia.

Hypermobility

Hypermobility is defined as excessive joint movement within a normal plane of motion.

The term hypermobility is frequently used interchangeably with the terms hyperextensibility and hyperlaxity, although technically the latter refers to movement in abnormal planes (Coles et al., 2017). Hypermobility can be hereditary, constitutional, structural, or the result of functional changes in the joint and surrounding tissues. Hypermobility can be localized or generalized. Localized hypermobility may be due to trauma, such as a joint dislocation, surgery, training (i.e. excessive stretching in gymnastics), but can also be hereditary. Generalized or systemic hypermobility, involving more than five joints, is usually hereditary with individual differences based on age, sex, and ethnic background (Coles et al., 2017). Systemic hypermobility is often assessed with the Beighton Score (Figure 7.1), which has acceptable inter-rater reliability but inconclusive validity (Juul-Kristensen et al., 2017). Several other commonly used tests lack satisfactory reliability and validity (Juul-Kristensen et al., 2017). The Five-Part Questionnaire (Table 7.1) is the most commonly used questionnaire for adults with conflicting evidence of reliability and validity (Juul-Kristensen et al., 2017; Hakim and Grahame, 2003). The Bristol Impact of Hypermobility test, which aims to measure the impact of hypermobility on a person's life, showed excellent test-retest reliability (Palmer et al., 2017).

Nearly all children could be considered hypermobile with girls having greater joint mobility than boys. Range of motion tends to increase until adolescence then tapers off in adulthood and old age. Because joint mobility in children is far greater than in adults, diagnosing hypermobility in children is challenging as there are no age-related norms for range of motion in children. An Australian study of 1584 subjects showed that 61% of girls and 37% of boys were hypermobile (Morris et al., 2017), which does not necessarily indicate

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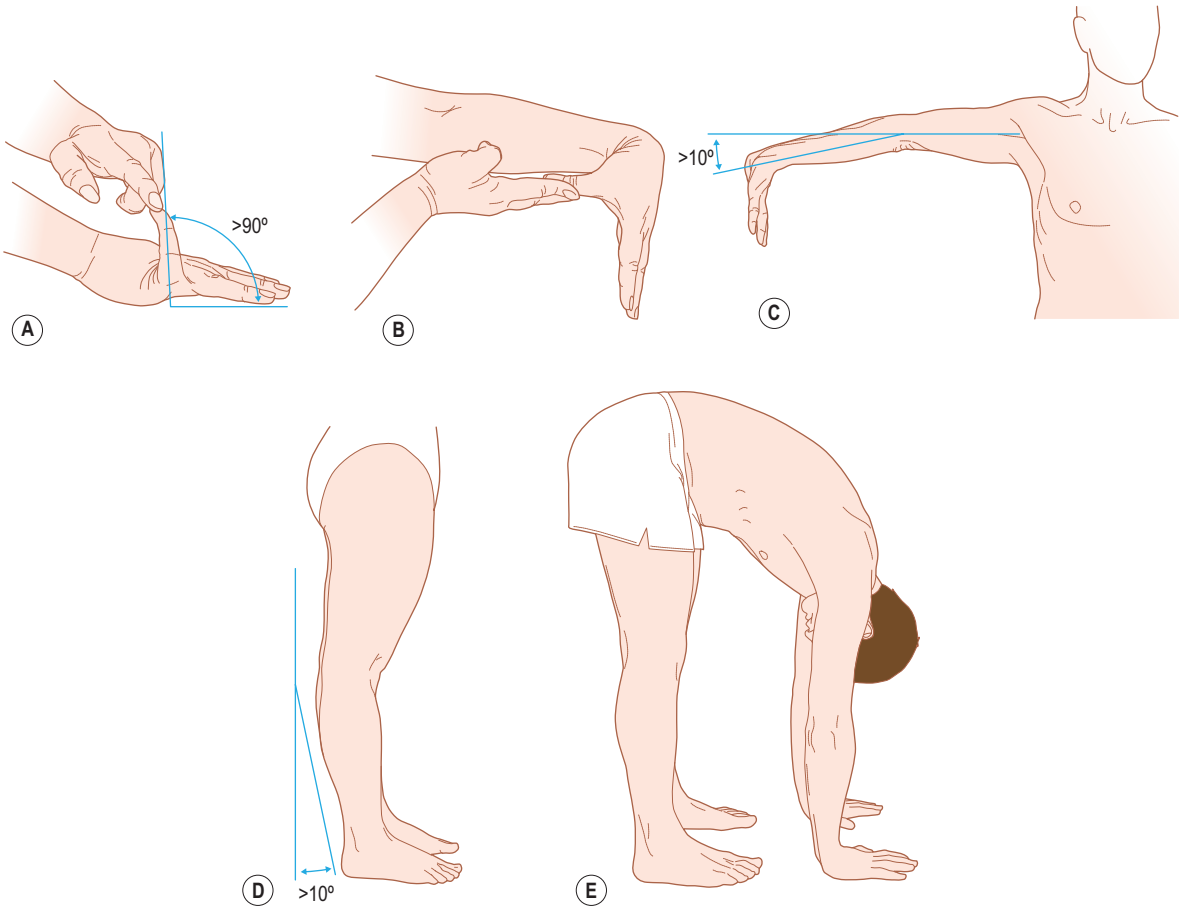


Figure 7.1 The Beighton tests

Tests A–D are performed bilaterally and one point is awarded for each positive test for a possible total of 0–8. Test E is either positive or negative; one point is awarded for a positive outcome. Although there is no universal agreement on the interpretation of the Beighton tests, many researchers agree that a score of 4–9 is indicative of systemic hypermobility.

Table 7.1 The Five-Point Questionnaire (Hakim and Grahame, 2003)

1. Can you now (or could you ever) place your hands flat on the floor without bending your knees?
2. Can you now (or could you ever) bend your thumb to touch your forearm?
3. As a child, did you amuse your friends by contorting your body into strange shapes or could you do the splits?
4. As a child or teenager, did your shoulder or kneecap dislocate on more than one occasion?
5. Do you consider yourself “double-jointed”?

A “yes” answer to two or more questions suggests joint hypermobility with 80–85% sensitivity and 80–90% specificity.

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that these children had abnormal range of motion or that they experienced disproportionate pain or dysfunction. Javadi Parvaneh and Shiari (2016) modified the Beighton criteria to better identify hypermobility in children.

Pain is a very common feature of hypermobility syndromes, both in children and adults, but not all hypermobile people are necessarily symptomatic (Juul-Kristensen et al., 2017). A study of 466 hypermobile adults reported that 99% suffered from joint pain and 91% from extremity pain along with several other symptoms including chronic fatigue (82%), anxiety (73%) and depression (69%) (Murray et al., 2013). Bettini et al. (2018) concluded that joint hypermobility was a common precursor to pain hypersensitivity and central sensitization in 40 hypermobile adolescents. Managing pain in hypermobile patients can be challenging and often requires a multimodal interdisciplinary approach (Revivo et al., 2019; Zhou et al., 2018). Pain management may include manual therapy, such as trigger point therapy and soft tissue mobilizations (Tewari et al., 2017), pain science education (Louw, 2016), cognitive, emotional and behavioral therapy (Baeza-Velasco et al., 2018), and external-focus exercise therapy (Wulf and Lewthwaite, 2016; Wulf et al., 2018). Although there is little known about yoga, Pilates and other bodywork in the context of painful hypermobility syndromes, there is no question that these approaches can be very valuable for this patient population. For example, yoga has been shown to be more effective in reducing chronic low back pain compared to conventional exercise classes, a self-care book and standard medical care (Williams et al., 2003; 2009; Sherman et al., 2005). Yoga instructors may have to modify the exercises and poses to stay within a pain-free range, while keeping the focus on improving the client's whole-body and individual joint stability, muscle strength and endurance, and improved

proprioception. In addition, yoga may be able to reduce anxiety and stress by incorporating a more mental focus with breathing exercises, such as pranayama, and meditation (West et al., 2004; Kabat-Zinn, 1982). Pilates training has also been shown to improve mental health outcomes (Fleming and Herring, 2018).

Asymptomatic hypermobility can be an asset for certain activities, like yoga (Nilsson et al., 1993), ballet (Foley and Bird, 2013), modern dance (Ruemper and Watkins, 2012), gymnastics (Bukva et al., 2019), acrobatics (Purnell et al., 2010), figure skating (Tosi et al., 2019) and cheerleading (Shields and Smith, 2011). Studies found that 44–58% of dancers were systematically hypermobile (Day et al., 2011; Sanches et al., 2015). A study of elite dancers determined that over 70% met the Beighton Score criteria (Chan et al., 2018). In general, hypermobility does not necessarily increase the risk of injury (Bukva et al., 2019), although several studies did find a correlation between joint hypermobility and risk of injury (Ruemper and Watkins, 2012; McCormack et al., 2004; Tosi et al., 2019). Fifty percent of acrobatic gymnasts reported pain in their spine, hips and knees (Anwajler et al., 2005). A study of Australian yoga students did not identify any evidence of an increased risk of injury (Penman et al., 2012), but other studies revealed an increase in pain in participants with degenerative joint disease and facet arthropathy (Lee et al., 2019). The authors cautioned against extreme spinal flexion for individuals with osteopenia or osteoporosis due to a higher risk of compression fractures or deformities.

Whether hypermobility is an asset or a liability in these activities continues to be an issue of debate (Foley and Bird, 2013; Grahame and Jenkins, 1972). On one hand, being hypermobile may make such activities easier, but on the other hand, it is conceivable that the risk of micro-trauma and

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joint instability may increase (Harris et al., 2016; Weber et al., 2015, Ruemper and Watkins; 2012). In general, the benefits of yoga, dance and other physical activity, such as improved cardiovascular health, increased muscle strength, endurance and control, may outweigh the risks of injury (Hayes and Chase, 2010; Wiese et al., 2019). Contact sports do appear to increase the risk of injury in hypermobile athletes (Rejeb et al., 2019).

Inherent joint laxity may have directed individuals towards dance and gymnastics (Nilsson et al., 1993), but for others, increased joint laxity may be the result of regular training involving excessive stretching. Individuals with significant joint laxity, Marfan's syndrome, or Ehlers-Danlos syndrome, should be screened medically for cardiovascular abnormalities (i.e. fluctuating blood pressures, postural orthostatic tachycardia syndrome and mitral valve prolapse) and soft tissue dysfunction (i.e. tissue fragility and potential aneurysms) prior to engaging in physically challenging activities (Grahame, 1993). The focus of yoga, dance and gymnastics should shift towards maintaining and improving joint and overall body stability in symptomatic hypermobility and not emphasize flexibility. When hypermobile individuals engage in stretching exercises, typically they primarily stretch their joint capsules and ligaments and not their muscles and tendons, which will potentially increase their hypermobility and risk of subluxations and dislocations (Shields and Smith, 2011).

Hypermobile individuals can be divided into three broad groups: (1) asymptomatic hypermobility, (2) syndromic joint hypermobility, and (3) hypermobility spectrum disorders (Coles et al., 2017).

Syndromic joint hypermobility

Most hypermobility syndromes are hereditary disorders, such as the Ehlers-Danlos Syndromes

(EDS) (Malfait et al., 2017), Marfan Syndrome (MFS) (Loeys et al., 2010) and Osteogenesis Imperfecta (OI) (Van Dijk et al., 2010), among others. EDS is the most common inherited connective tissue disorder with approximately 1 in 5000 births (Tinkle et al., 2017), compared with 1 in 3000–10 000 for MFS (Pyeritz, 2017) and 1 in 10 000–15 000 for OI (Lafage-Proust and Courtois, 2019; Morello, 2018).

Ehlers-Danlos Syndromes

According to the updated EDS clinical classification, published in 2017, Ehlers-Danlos Syndromes are a heterogeneous group of 13 different but overlapping connective tissue disorders, featuring joint hypermobility, skin hyperextensibility and fragile tissues (Table 7.2). Many patients with EDS experience persistent pain, autonomic dysfunction (dysautonomia) and gastrointestinal dysmotility (Beckers et al., 2017). Each subtype has its own specific major and minor criteria. For all subtypes, except the hypermobile subtype, the definitive diagnosis must be made by molecular confirmation with identification of the involved gene(s), which would also determine the particular inheritance pattern, specific risk factors, and guide the therapeutic management.

As the genetic basis of hypermobile EDS is unknown, the diagnosis is made based on history, examination and a clinical impression (Malfait et al., 2017). The autosomal dominant classical (cEDS) and hypermobile (hEDS) types are the most common. The vast majority of cEDS patients have heterozygous mutations in one of the genes encoding type V collagen (COL5A1 and COL5A2) (Bowen et al., 2017). According to the new EDS criteria, the diagnoses of hEDS and joint hypermobility syndrome are no longer considered two separate entities. Instead, both diagnoses are included in the definition of hEDS, characterized not only by hypermobility, but also

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Table 7.2 Clinical classification of the Ehlers-Danlos Syndromes (EDS), genetic basis and protein type (Malfait et al., 2017)

Clinical subtype	Genetic bases	Proteins
Classical EDS	Major: COL5A1, COL5A1 Rare: COL1A1	Type V collagen Type I collagen
Classical-like EDS	TNXB	Tenascin XB
Cardiac-valvular EDS	COL1A2	Type I collagen
Vascular EDS	Major: COL3A1 Rare: COL1A1	Type III collagen Type I collagen
Hypermobile EDS	Unknown	Unknown
Arthrochalasia EDS	COL1A1, COL1A2	Type I collagen
Dermatosparaxis EDS	ADAMTS2	ADAMTS2
Kyphoscoliotic EDS	PLOD1 FKBP14	LH1 FKBP22
Brittle cornea syndrome	ZNF469 PRDM5	ZNF469 PRDM5
Spondylodysplastic EDS	B4GALT7 B3GALT6 SLC39A13	B4GalT7 B3GalT6 ZIP13
Musculocontractural EDS	CHST14 DSE	D4ST1 DSE
Myopathic EDS	COL12A1	Type XII collagen
Periodontal EDS	C1R C1S	C1r C1s

by chronic pain, dysautonomia, chronic fatigue, anxiety, and other associated symptoms. The hEDS type represents at least 80–90% of all EDS cases (Tinkle et al., 2017). Many patients with hEDS experience significant levels of disability, which is highly correlated with both physical and psychological factors (Scheper et al., 2016).

As part of the differential diagnostic process, patients must be screened for possible neurological complications resulting from tissue weakness, biophysical deformative stresses, entrapments, and tissue deformations (Henderson et al., 2017). Comorbid conditions reported

with hEDS include Chiari malformations with or without tethered cord syndrome (Milhorat et al., 2007) and craniocervical instability with or without ventral brainstem compression (Henderson et al., 2018). Other common comorbidities include mast cell activation syndrome (Seneviratne et al., 2017), gastrointestinal dysfunction (Botrus et al., 2018), and postural orthostatic tachycardia syndrome (POTS) (Bonamichi-Santos et al., 2018). Many hypermobile EDS patients can benefit greatly from adjunctive bodywork approaches, such as Pilates (McNeill et al., 2018; Lewitt et al., 2019). For a successful Pilates

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training program, the medical classification of EDS may not be as relevant as the actual observations of the implications of hypermobility on movement patterns and motor control (Fayh et al., 2018). A qualified Pilates instructor may want to focus on establishing neuromuscular connections with slow-motion coordinated movement patterns emphasizing coordination, balance, breathing, and proprioception (McNeill et al., 2018; Bradley and Esformes, 2014). Pilates can be very effective in reducing pain and disability (Byrnes et al., 2018). Of course, the same applies to other fields, such as dance, yoga, fitness and other movement therapies.

Marfan Syndrome

Marfan syndrome (MFS) is another autosomal dominant heritable disorder of connective tissue. According to the revised Ghent Nosology for Marfan Syndrome, the cardinal clinical features are aortic root aneurysm and ectopia lentis (Loeys et al., 2010). Mutations of TGF- β are also associated with Loeys-Dietz syndrome (De Carlo et al., 2018). Patients with MFS frequently experience sleep apnea associated with a narrow and high palate (Paoloni et al., 2018), cerebrovascular disease (Kim et al., 2018), thoracic aortic and other aneurysms (Faure et al., 2018). The health-related quality of life of MFS patients is below the population norm; socioeconomic factors, such as college education, marital status, higher household income, private health insurance, full-time employment and moderate alcohol use, were predictors of a better quality of life (Goldfinger et al., 2017).

Patients with MFS have a defect of the gene *FBN1* responsible for encoding fibrillin-1, one of the main proteins of the extracellular microfibril, which may increase transforming growth factor beta (TGF- β) activity and lead to dilatation and aortic dissection, mitral valve prolapse, excessive bone growth, pulmonary blebs and

spontaneous pneumothorax, and several other syndromes, including some forms of dwarfism. More than 1200 mutations have been identified (Pyeritz, 2017). Similar to EDS, bodywork approaches can be very helpful for patients with Marfan's Syndrome.

Osteogenesis Imperfecta

Osteogenesis Imperfecta, or brittle bone disease, is known for its genetic and clinical heterogeneity. It is an autosomal dominant inherited, generalized, connective-tissue disease with monoallelic mutations in type I collagen genes *COL1A1* and *COL1A2* (Reuter et al., 2013). Currently, there are 18 different types of OI based on possible genetic mutations ranging from mild (OI type 1), moderate (OI type 4), severe (OI type 3) to lethal (OI type 2) (Kang et al., 2017; Morello, 2018), but the list continues to expand (Lafage-Proust and Courtois, 2019). At least one recessive type is X-chromosome-linked. OI type 5 is due to a mutation of the *IFITM5* gene. OI involves mutations in type I collagen, type I collagen synthesis and the differentiation and function of osteoblasts. It is beyond the scope of this chapter to review the many genetic mutations involved in OI.

Patients with OI have a predisposition for fractures, short stature, bone deformities, ligamentous laxity, joint hypermobility, vascular fragility, ocular and dental abnormalities, respiratory impairments due to severe scoliosis, relative macrocephaly, blue sclerae, the presence of sutural and intrasutural bones on skull radiography, progressive post-pubertal hearing loss, insertional tendinopathies, craniovertebral junction abnormalities (mostly in OI types 3 and 4), including platybasia, basilar invagination and basilar impression. Adults with unexplained bone fragility should be examined for OI, MFS, Gaucher disease, hypophosphatasia and EDS.

Hypo- and hypermobility

Treatments with bisphosphonates, such as neridronate and risedronate, teriparatide, anti-sclerostin antibody setrusumab and denosumab have not reduced the fracture risk, but they may increase bone density (Lafage-Proust and Courtois, 2019). Orthopedic fracture management is challenging, due to the craniovertebral junction abnormalities, poor bone structures, bleeding tendencies, and post-operative respiratory failure. The overall management should focus on individualized multidisciplinary care. Physiotherapy and bodywork should include treatments to strengthen muscles, reduce pain, treat tendinopathies, and to facilitate movement and optimal gait.

Hypermobility spectrum disorders

With the development of the stricter criteria for EDS, several patients with hypermobility

will not meet any of the new criteria (Malfait et al., 2017) and for these individuals the term hypermobility spectrum disorders (HSDs) was coined. They usually have musculoskeletal symptoms, although some may have limited multisystem involvement. Both hEDS and HSD patients feature a myofibroblast-like phenotype with several abnormal extracellular matrix components (ECM), such as different expressions of CCN1/CYR61 and CCN2/CTGF inflammation mediators, an altered organization of α smooth muscle actin cytoskeleton, and increased levels of the ECM-degrading metallo-proteinase-9, among others (Zoppi et al., 2018). HSDs are divided into generalized (G-HSD), peripheral (P-HSD), localized (L-HSD) and historical (H-HSD) subtypes. The genetic basis, prevalence and incidence of HSDs are not known.

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Hypo- and hypermobility

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